

DOCUMENT NNV-RL | REV A | 16 September 2026

Reading list

Annotated: where each idea in the session came from

Start here

- **The Vehicle tutorial.** Chapters: introduction; the language (ACAS Xu); proving robustness (MNIST); property-driven training; exporting to theorem provers; *Vancomycert*. The hands-on exercise is adapted from chapters 2–4. <https://vehicle-lang.github.io/tutorial/>
- **Tutorial chapter 4, property-driven training** — a Fashion-MNIST classifier at 99.9 % training accuracy has only 21–25 of 50 images provably robust at $\varepsilon = 0.005$; accuracy is not safety. <https://vehicle-lang.github.io/tutorial/chapters/property-driven-training/>
- **Vehicle documentation and source** — reference docs <https://vehicle-lang.readthedocs.io/en/latest/>; compiler (BSD-3-Clause) <https://github.com/vehicle-lang/vehicle>; tutorial repository, branch exercises, <https://github.com/vehicle-lang/tutorial>
- **Tutorial provenance** — run at FoMLAS@CAV 2023, the VeTSS Summer School 2023, ICFP and the SAT/SMT Summer School; tutorial paper on EasyChair: <https://easychair.org/publications/download/Rkrv>
- Cordeiro, Daggett, Girard-Satabin, Isac, Johnson, Katz, Komendantskaya, Lemesle, Manino, Šinkarovs, Wu. *Neural Network Verification is a Programming Language Challenge*. ESOP 2025. arXiv:2501.05867 — why specification languages, and the embedding gap.
- Kaulen, Ladner, Bak, Brix, Duong, Flinkow, Johnson, Koller, Manino, Nguyen, Wu. *The 6th International Verification of Neural Networks Competition (VNN-COMP 2025)*. arXiv:2512.19007 — the ecosystem: ONNX + VNN-LIB, eight tools, 25 benchmarks.

Verification: foundations and tools

- Katz, Barrett, Dill, Julian, Kochenderfer. *Reluplex: An Efficient SMT Solver for Verifying Deep Neural Networks*. CAV 2017. arXiv:1702.01135 — the ACAS Xu benchmark, NP-completeness, and the case-splitting idea behind Marabou.
- Wu et al. *Marabou 2.0: A Versatile Formal Analyzer of Neural Networks*. arXiv:2401.14461 — the solver used today.
- Daggett, Kokke, Atkey, Arnaboldi, Slusarz, Komendantskaya et al. *Vehicle: Interfacing Neural Network Verifiers with Interactive Theorem Provers*. arXiv:2202.05207; and *Bridging the Embedding Gap in the Verification of Neuro-Symbolic Programs*, FSCD 2025 (arXiv:2401.06379) — the car controller that provably stays on the road.
- Athavale et al. *Verifying Global Two-Safety Properties in Neural Networks with Confidence*. CAV 2024 — self-composition (the twin network) for monotonicity and fairness.

Adversarial examples

- Szegedy et al. *Intriguing properties of neural networks*. 2013. arXiv:1312.6199 — the first adversarial examples: imperceptible noise flips the label.

- Goodfellow, Shlens, Szegedy. *Explaining and Harnessing Adversarial Examples*. 2014. arXiv:1412.6572 — the panda that became a gibbon.
- Eykholt et al. *Robust Physical-World Attacks on Deep Learning Visual Classification* (CVPR 2018; arXiv:1707.08945) — stickers on a stop sign read as a speed limit.
- *Revisiting Physical-World Adversarial Attack on Traffic Sign Recognition: A Commercial Systems Perspective*. NDSS 2025. arXiv:2409.09860 — the stickers still work on commercial systems.
- Finlayson et al. *Adversarial attacks on medical machine learning*. *Science* 363:1287, 2019 — medical images are attackable too.
- Sharif et al. *Accessorize to a Crime*. CCS 2016 — adversarial eyeglass frames make you someone else to a face recogniser.

Medical

- Sirman, Conway, Ciupa, Grinbergs, Komendantskaya, Hoang, Rawson, Bruni, Belle, Williams. *Vancomycert: A Certified Neuro-Symbolic Drug Delivery System*. SAIV 2026. arXiv:2606.19532.
- Royal College of Physicians. *National Early Warning Score (NEWS) 2*. 2017 — the labelling rule behind the triage exercise (Charts 1 and 4). <https://www.rcp.ac.uk/improving-care/resources/national-early-warning-score-news-2/>
- Wong et al. *External Validation of a Widely Implemented Proprietary Sepsis Prediction Model*. *JAMA Internal Medicine*, 2021 — deployment on vendor validation alone.
- *AI-Enabled Medical Devices Drive Nearly Half of All Recalls*. Biomedical Safety & Standards, Feb 2026.

Incidents cited

- Waymo recall after the San Antonio flood incident (20 Apr 2026): CNBC, 12 May 2026; SF Chronicle; Texas Public Radio; Electrek. NHTSA upgrades the Tesla FSD probe (EA26002) to an engineering analysis: *Insurance Journal*, 20 Mar 2026.
- Met Police live facial recognition: High Court judgment of 21 Apr 2026 (The Register, 22 Apr 2026); Big Brother Watch press release; OECD.AI incident record 2026-04-21; 2025 deployment figures as reported by PA in 2026 (not yet checked against the Met's own statistics).
- PocketOS production database deleted by a coding agent (Apr 2026): ABC7; ACS Information Age ("Gone in 9 seconds"); Zenity. Forbes Tech Council, *When an AI agent goes rogue, who takes the blame?*, 29 Jun 2026.
- AI Incident Database roundups, 2026: <https://incidentdatabase.ai/blog/>

Regulation

- EU AI Act, Article 15 (accuracy, robustness and cybersecurity) and Annex III (high-risk systems, incl. credit scoring): <https://artificialintelligenceact.eu/article/15/>. High-risk obligations apply from 2 August 2026.
- UK Equality Act 2010 (indirect discrimination): <https://www.legislation.gov.uk/ukpga/2010/15/contents>